Sample paper

Note: The end-semester exam will have two types of questions, and one code snippet to write the output.
Another is to complete the code snippet for the given problem.

Q1. Write the output of the given snippet code. Also, give the trace of output.

```
struct Ournode{

char x,y,z;

};

int main(){

struct Ournode p = {'1', '0', 'a'+2};

struct Ournode *q = &p;
printf ("%c, %c", *((char*)q+1),
*((char*)q+2));

return 0;

}
```

here q point the x
// *(8x+1) = '0'
= *(8x+2) = 'c'

// here x = '1'
y = '0'
z = 'q'+2
As ASCII q = 97 / 97+2 = 99 = c
z = 'c'

Output = 0, c

Q2 Write a recursive function to calculate the factorial of a number.
Enter a number to calculate its factorial: 5
Factorial of 5 is: 120

```
/*complete the function*/
int factorial(int n)
{
  if (n==0||n==1)
      return 1;
  else
     return n * factorial(n-1);



}

int main()
{
        int number;

        printf("Enter a number to calculate
its factorial: ");
        scanf("%d", &number);


        if (number < 0)
        {
        printf("Factorial of a negative
number is not defined.\n");
        }
        else
        {

        printf("Factorial of %d is: %d\n",
number, factorial(number));
        }

        return 0;
}
```